

PSA for Hermès Bags: Decision-Grade Trust Infrastructure Memo + Build Plan for HSA

Executive summary

0) Executive decision: Proceed with constraints — run a tightly-scoped 12–18 month build with explicit kill criteria and a trust-first operating model (authentication + rating/grading + guarantee + chain-of-custody artifact), not a GMV-first marketplace.

What matters (and what must be true)

HSA can be “the PSA of Hermès” only if it becomes the *reference standard* for (i) **authenticity determination**, and (ii) **auditable, repeatable condition/originality grading** that becomes legible to buyers, resellers, insurers, and marketplaces. PSA’s defensibility comes from (a) a widely accepted numeric grade signal, (b) a tamper-evident physical artifact bound to a database-verified certificate number, and (c) a financial guarantee + error-correction machinery that market participants can underwrite into pricing behavior. PSA explicitly commits to grading “under PSA grading standards,” uses “two or more... graders” who “must agree upon a grade,” and offers a buyback/refund mechanism if a grade is later determined incorrect. ¹

HSA must replicate those *trust mechanics* for Hermès bags, but with redesigns for (1) high-value, repairable goods where “state changes” (spa/repair, component swaps) are common, and (2) an adversarial environment where simple certificates and naïve NFC tags fail quickly.

Biggest risks (adversarial framing)

The primary failure mode is not demand. It is *standard failure*: inconsistent grading, tag decoupling (certificate → bag mismatch), or guarantee blow-ups that collapse trust. Secondary risks are brand/IP pressure (luxury tends to litigate more aggressively than hobby collectibles), and counterfeit adaptation that targets HSA’s weakest link (usually chain-of-custody and re-sale verification surfaces).

Key numbers (Verified vs Estimated vs Unknown)

Verified (primary sources): - PSA/Collectors Universe FY2020 performance: net revenues **\$78.891M**, gross profit **\$45.236M** (gross margin ~57.3%), operating income **\$14.102M** (operating margin ~17.9%). ²

- **Collectors Universe segment profitability (FY2020):** Coins generated **\$41.599M** net revenues with **\$11.995M** segment operating income; Trading Cards & Autographs generated **\$33.673M** net revenues with **\$9.328M** segment operating income. ²

- **PSA demand throttle mechanics:** higher fees for faster turnaround are a margin driver (mix shifts to faster service increases ASP). ³

- **PSA guarantee mechanics (baseline analog for HSA):** If PSA later concludes a card no longer merits the assigned grade or fails authenticity standards, PSA will **buy the card** (in some cases) or **refund the difference in value** if the grade is lowered. ¹

- **PSA current service-tier pricing (illustrative ASP range):** e.g., Value Plus **\$44.99/card**, Regular **\$74.99**, Express **\$149.99**, Super Express **\$299.99**, Walk-Through **\$599.99**, plus higher service levels. ⁴

- **Entrupy Hermès premium unit price (competitive reference):** Hermès handbags/clutches priced **\$119**

per authentication, results “within 24 hours” (Entrupy pricing page). ⁵

- **StockX promise scope (authenticity + condition gate):** StockX states items are “**Verified by StockX**” (not endorsed by brands) and accepts handbags only in “**Brand new**” condition (includes required packaging/accessories). ⁶

- **StockX Buyer Promise dispute window:** buyer must notify StockX “**within 10 days**” of receipt; if a “Verified Authentic” tag was attached, “**the original tag must remain attached**” for a claim. ⁷

- **StockX trust controversy matters because it becomes court-testable:** In *Nike, Inc. v. StockX LLC*, the court order (Mar 4, 2025) granted partial summary judgment that StockX is liable for distributing counterfeit goods re: **4 pairs** sold to Nike investigators and **33 pairs** sold to Roy Kim. ⁸

- **Why naïve NFC fails, in one line:** academic work notes that NFC Forum does not specify a tag authentication mechanism; absent that, NFC tags “cannot be used for secure product identification,” and cloning is a core concern. ⁹

- **Minimum secure NFC direction (what “good” looks like):** NXP’s NTAG 424 DNA describes AES-128 cryptography generating a tap-unique Secure Unique NFC (SUN) message; the phone reads a “tap-unique URL,” sends it to a host, and the host verifies authenticity. ¹⁰

Estimated (explicit assumptions; sensitivity required): - HSA **Grade + Seal** target price: **\$550–\$850/bag** depending on declared value and turnaround (estimate; see C3 model).

- HSA **fully-loaded cost per graded bag** (labor + overhead + materials + dispute reserve): **\$180–\$320/bag** (estimate; sensitivity in C3).

- HSA **launch dispute rate:** **0.4%–1.0%** within 90 days (estimate; must be empirically validated via pilots).

- HSA **break-even** (single hub): **~300–800 bags/month** depending on wage geography + fixed footprint (estimate; see C3).

Unknown (must be discovered during diligence/pilot): - True counterfeit base rate in Hermès resale submissions by region + channel; true false-negative severity distribution (tail risk).

- Brand enforcement posture toward a third-party “grading” standard for handbags in target jurisdictions.

- Long-run willingness-to-pay for *graded condition*, not just authenticity, among top Hermès dealers.

Falsification checklist (30–60 day tests; pass/fail thresholds)

These are designed as *kill switches*, not vanity KPIs:

1. **Inter-rater reliability:** on **n=120 bags** (balanced across condition tiers), achieve **≥85% agreement** within **±1 condition notch** and **≥70% exact-match** on final condition tier after protocol training; **fail** if <75% within ±1.
2. **Counterfeit catch-rate (seeded red team):** insert **n=40 known counterfeits/superfakes** (sourced legally for testing) across categories; **pass** if **0** are certified authentic and **≤5%** become “inconclusive”; **fail** if **≥1** passes as authentic.
3. **Tag security (lab attack):** attempt clone/replay/tag transplant on **n=30 tags**; **pass** if successful counterfeit verification is **0%** under documented attacker models; **fail** if any attack yields a “valid” verification state.
4. **Certificate-to-bag binding audit:** for **n=200 certified bags**, randomly re-run verification at 2 weeks; **pass** if **≥99.5%** resolve to correct bag identity + state; **fail** if any systematic mismatch appears.
5. **Dispute handling SLA:** resolve **≥90%** of disputes within **10 business days**; **fail** if median >15 business days (trust corrosion).
6. **Dealer adoption (standard credibility):** sign **≥6** high-velocity Hermès resellers to written “use-grade-in-listing” commitments (or API integration LOIs); **fail** if <3.

7. **Price premium evidence:** blind A/B marketplace test with partner: same bag listings with/without HSA grade; **pass** if HSA-grade listings show $\geq 10\%$ **higher conversion** or $\geq 5\%$ **higher net realized price** at constant exposure; **fail** if no measurable effect.
 8. **Loss ratio ceiling:** in pilot, keep total guarantee payouts + refunds $\leq 0.5\%$ of **certified declared value**; **fail** if $> 1.0\%$ (unsustainable guarantee economics).
 9. **Throughput feasibility:** hit stable run-rate of ≥ 6 **fully graded bags/authenticator/day** (net of QC and rework) by week 8; **fail** if < 4 without clear fix.
 10. **Brand/IP risk scan:** obtain counsel memo + monitoring: **no** credible C&D with injunctive threat in first 60 days; **fail** if litigation risk becomes immediate and existential.
-

Definitions & taxonomy

Definitions & taxonomy (required)

Authentication = a determination that an item is genuine (or not) based on evidence and expert judgment; it is not “title.” eBay explicitly distinguishes that its authenticity certificate is a “tool” and “does not represent title to the physical good.” ¹¹

Grading = a standardized evaluation of physical state/quality (often numeric) intended to be comparable across items; PSA grades on a 1–10 scale and positions its grade as a market signal. ¹²

Rating = a categorical or tiered condition signal (often not as granular as a formal grade).

Certificate = a verifiable record asserting authentication and/or grade; for resilience it must be bound to an immutable identifier and ideally to evidence (photos, measurements). Entrupy certificates are hosted on Entrupy’s servers and meant to be validated by entering the certificate link. ¹³

Chain-of-custody = documented control states describing when the certifier had physical possession vs when the item is “out of custody,” which changes what can be guaranteed.

Tamper-evident = physical design where attempts to open/alter leave evidence; PSA describes holder tamper evidence via “frosting” and changes in plastic rigidity after sonic weld violation. ¹⁴

Registry = a database of certified items and their attributes; PSA provides certification verification and encourages public cert checks. ¹⁵

Population report = aggregate data describing how many items exist in each grade; PSA markets its Population Report as a core collector resource. ¹⁶

SLA = a service level agreement defining turnaround + dispute handling, essential because PSA and others explicitly warn turnaround times are estimates and can change with capacity. ¹⁷

Loss ratio = for HSA, payouts + refunds + dispute costs divided by revenue or declared value; must be bounded for a viable guarantee product.

Chargeback = payments reversed by card networks; in trust services, chargebacks correlate with disputed authenticity/condition.

False positive / false negative = for authentication: - false positive = authentic item rejected (customer pain, reputational cost)

- false negative = counterfeit certified authentic (existential trust failure)

Secure element = hardware designed to store cryptographic secrets and execute secure operations.

Cryptographic NFC, challenge-response, key management = systems where tags prove authenticity via cryptographic operations, not static identifiers. NXP's NTAG 424 DNA describes AES-based SUN messages and backend verification, which is the direction HSA must use. ¹⁰

Part A — PSA Case Study

A1) PSA history & timeline (concise but complete)

Timeline table (mechanism-first)

Date	Event	Why it mattered (mechanism)	Evidence
1986	PCGS founded	Establishes slab-based grading + trust artifact pattern later mirrored in other categories	Collectors Universe 10-K notes PCGS founded 1986. ²
1991	PSA founded	Creates independent third-party grading standard for cards; slab + population/registry loop becomes price signal	Collectors Universe 10-K notes PSA founded 1991. cite turn3view0 
2001	PSA Set Registry opened (2002 awards)	Formalizes competition + status incentives; strengthens network effects and switching costs	PSA "Set Registry Started in 2001... Awards... 2002."
FY2020 (June 30, 2020)	\$78.9M revenue with ~57% gross margin	Demonstrates scaled unit economics of trust services (labor-heavy but high GM with operating leverage)	Collectors Universe FY2020 financials. ²
Sep 30, 2020	Record backlog (cards/autographs); ASP increases via faster turnaround tiers	Shows demand shocks and how tiered pricing/turnaround acts as throttle; backlog is a moat <i>only if</i> trust remains intact	Collectors Universe 10-Q discusses record backlog and higher ASP from faster service. ¹⁸

Date	Event	Why it mattered (mechanism)	Evidence
Nov 30, 2020	Take-private announced: \$75.25/share, ~\$700M	Ownership shift enables capital + capacity expansion; also signals strategic value of trust infrastructure	Company press release via GlobeNewswire. ¹⁹
Feb 22, 2021	Acquisition completed	Confirms post-public ownership and expansion pathway	Business Wire announcement. ²⁰
Dec 2, 2021	Collectors acquires Card Ladder	Strengthens pricing-data rails and potentially accelerates standard propagation (pricing + registry complements grading trust)	PSA/Collectors article on Card Ladder acquisition. ²¹
May 16, 2024	Collectors + eBay transactions close (vault/flow integration)	Distribution: embeds grading into transaction flow, making the standard “default” behavior	PR Newswire release. ²²

Corporate structure & leadership / ownership shifts

Collectors Universe (public) historically owned PSA and PCGS; it disclosed segment reporting across Coins, Trading Cards & Autographs, and “Other,” showing authentication is the core revenue engine. ²³

The investor group led by Nat Turner completed acquisition in Feb 2021 (take-private). ²⁰

Collectors’ current leadership page lists Nat Turner as Chairman & CEO and Steve Cohen among board members, and lists brands including PSA, PSA Vault, PCGS, Beckett, SGC, and Card Ladder. ²⁴

Inflection points that plausibly form today’s moat (mechanism-linked)

First, PSA’s Set Registry (2001) converts grading from “quality label” into a status game with durable network effects: collectors compete on registry sets, increasing willingness to pay for grading and decreasing willingness to switch standards.

Second, guarantee + error-correction policy is not marketing fluff; PSA explicitly promises it will buy a card or refund grade-value differences if PSA later determines grade/authenticity issues under its standards. That converts the grade into an insurable economic claim. ¹

Third, physical tamper evidence + database verification reduces fraud surface: PSA emphasizes sonic-weld tamper evidence and public cert verification pathways. ²⁵

Fourth, pricing/turnaround tiers + backlog dynamics create a capacity moat: SEC filings explain that faster service commands higher fees and that a portion of cost of revenues is relatively fixed short-term, enabling operating leverage at high volumes—if trust survives. ²⁶

Fifth, distribution integrations (eBay, dealer programs) spread the standard by embedding authentication into transaction defaults; PSA describes eBay checkout integration where PSA authenticates the card and verifies condition matches listing before grading. ²⁷

A2) PSA success factors (micro + macro)

Micro/internal mechanisms

Trust formation mechanics (why the market accepted PSA)

PSA's trust proposition is a bundle: (i) a standard, (ii) process controls, and (iii) financial responsibility. The FAQ makes two operationally critical claims: *multi-rater review* ("two or more... graders... must agree") and *economic accountability* (buyback/refund policy). ¹

That combination changes market behavior: it makes "PSA grade" a tradable signal and gives counterparties a common language for sight-unseen trade.

Network effects: registry + population report + adoption loops

PSA's Population Report provides public supply distribution by grade, which becomes price discovery infrastructure (scarcity + liquidity). ¹⁶

The Set Registry creates a structured competition that rewards grading participation and drives repeat submissions.

The loop is: more submissions → richer population data + more registry competition → greater perceived legitimacy of PSA grades → more marketplace quoting in PSA terms → more submissions.

Pricing + capacity control: tiers/turnaround as demand throttle

Collectors Universe filings are explicit: turnaround times requested affect revenue because the company "charge[s] higher fees for faster service times," and higher ASP in a quarter came from customers paying for improved turnaround. ¹⁸

PSA's public service-tier menu makes the same strategy legible to customers and enables continuous repricing to match volume to capacity. ²⁸

Workflow/QA signals (what can be verified publicly)

While PSA does not fully disclose internal SOPs, it does make verifiable claims about:

- Certification verification via app/QR and cert lookup. ²⁹
- Reholder workflow that includes authenticity checks on the holder and card before re-encapsulation, implying chain-of-custody thinking and anti-counterfeit measures even for the holder itself. ¹⁷

Failure modes and mitigations

Backlog can destroy trust if it incentivizes corner-cutting. Collectors Universe disclosed a "record backlog" of card submissions as of Sept 30, 2020. ¹⁸

PSA's control lever is *pricing and service mix* (charge more for speed), but the unsolved piece is that grading remains labor-intensive and thus exposed to training drift, fatigue, and incentive distortion. PSA's multi-grader agreement rule is one mitigation, but HSA must design deeper auditability given higher claim severity for bags. ¹

Macro/external mechanisms

Market structure shift and why third-party standards become necessary

Collectors Universe describes authentication/grading as facilitating sales because it provides "a high level of assurance as to the authenticity and quality" of items being bought/sold, linking demand for grading to transactional behavior. ¹⁸

This is the macro logic HSA must exploit: as Hermès resale transactions become increasingly remote and cross-border, third-party standards become a prerequisite to liquidity.

Fraud dynamics and psychology

A credible standard is partly a coordination solution: it tells *both* parties what evidence counts and how disputes are resolved. Where disputes are economically meaningful, the guarantee and dispute process matter as much as the initial authentication.

Manufacturer relationship

PSA has historically operated without manufacturer endorsement, and eBay explicitly states its authenticity services are not endorsed by brands. ³⁰

For HSA, “not brand-endorsed” must be positioned as independence, not antagonism; but brand/IP risk is structurally higher in luxury (see legal posture later).

A3) PSA unit economics & financial reality (SEC-grounded, then translated)

SEC-extracted financials (Collectors Universe; FY ended June 30)

All numbers are from Collectors Universe filings prior to privatization.

Metric	FY2020	FY2019	Source / label
Net revenues	\$78.891M	\$68.449M	Verified (SEC). ²
Cost of revenues	\$33.655M	\$28.694M	Verified (SEC). ²
Gross profit	\$45.236M	\$39.755M	Verified (SEC). ²
Gross margin	~57.3%	~58.1%	Calculated from SEC values (verified inputs). ²
Operating income	\$14.102M	\$12.124M	Verified (SEC). ²
Operating margin	~17.9%	~17.7%	Calculated from SEC values (verified inputs). ²

Segment economics (why “grading” can be a high-margin labor business)

Collectors Universe segment reporting shows both Coins and Trading Cards & Autographs produced ~27–29% segment operating income margin before “unallocated” expenses. ²

This implies the core grading operations have meaningful operating leverage once throughput is high and rework/claims stay bounded.

Cost drivers (what actually eats COGS)

Collectors Universe (10-Q) explicitly enumerates major cost of revenues for authentication/grading: labor, production costs, credit card fees, warranty expense, and occupancy/security/insurance. ¹⁸

This is the template for HSA's cost model: labor is dominant; facility/security/insurance are meaningful fixed costs; warranty/guarantee is a tail-risk term that must be priced, not wished away.

Pricing architecture as an economic throttle (important for HSA design)

PSA's public menus show multi-tier pricing with declared value/insured value and turnaround times, and the Reholder page explicitly warns PSA may adjust pricing/services/processing time to align demand with capacity. ³¹

Collectors Universe filings reinforce the causal mechanism: faster turnaround commands higher fees;

vintage/higher-value items justify higher ASPs; and a portion of costs are fixed in the short term, so volume drives gross margin. ¹⁸

Indicative per-item economics (what is known vs unknown)

PSA does not publicly disclose annual units graded or per-item labor minutes in SEC filings (Unknown). Therefore, any direct “COGS per card” number is an estimate and should not be treated as verified.

What *can* be grounded: - **Price per item by tier** is public via PSA service pages. ²⁸

- **Cost categories** are SEC-grounded. ¹⁸

- Therefore, HSA's per-bag model (C3) should be built bottom-up, and compared to PSA's achieved gross margins as a plausibility bound, not as a copy-paste assumption. ²³

Part B — Pokémon card market vs Hermès bag market, plus comparables

B1) Similarities/differences that matter for HSA (evidence-based where possible)

Where PSA transfers cleanly (similarities)

Third-party certification becomes economically valuable when it enables remote trade by providing “assurance as to authenticity and quality” that facilitates buying/selling behavior. That is explicitly how Collectors Universe describes the driver of grading demand. ¹⁸

Hermès resale has the same structure: high price dispersion, asymmetric information, and high fraud payoff.

Where Hermès breaks the PSA model (differences that force redesign)

Repairability and “state changes” are endemic

Unlike cards in slabs, bags are living objects: spa treatments, re-dyeing, hardware swaps, strap replacement, and refurbishment can change both condition and authenticity evidence. Chinese reporting highlights a growing “旧包改造” (old bag modification) business, with shops accepting customer-provided bags and charging ~1000–3000 RMB depending on work; it also describes risks of “一眼假” outcomes and trademark issues. ³²

Implication: HSA must grade *originality state* (original vs repaired vs modified) and define re-auth triggers.

Luxury demand is value-preservation oriented (a driver of grading adoption)

Tencent Marketing Insight × BCG's 2024 report states China mainland consumers' luxury spending is estimated at **5,720亿元** in 2024 (+4%), and notes categories with stronger value-preservation attributes (珠宝、皮具) leading growth. ³³

Implication: Hermès buyers already reason in “value retention” terms—fertile ground for a credible grade signal if it is trusted.

Secondhand/grey-market complexity raises authentication stakes

A Reuters report describes China's growing second-hand and grey luxury markets, including discounts on platforms like DeWu (Dewu/得物) and estimates the grey market at ~\$57B annually, emphasizing price gaps

and consumer authenticity concerns. ³⁴
Dewu’s own product descriptions emphasize “先鉴别，后发货” (authenticate first, then ship) and claim “多道鉴别查验工序,” evidencing consumer demand for process-based trust (self-reported; treat cautiously). ³⁵

Standardizing “condition” is harder for bags than for cards
Cards have a constrained feature set (centering, corners, surface, edges). Bags have multi-material, multi-component condition surfaces and repairs that can be cosmetically invisible but value-relevant. eBay’s authenticity guarantee materials acknowledge there are cases where the authenticator may verify condition matches listing but “may not be able to verify... authenticity,” and will communicate what could not be authenticated. ³⁶
Implication: HSA must support an “inconclusive” state (and price it), and must make its grade claim precise.

- Evidence note (Chinese-language sources used in this section)**
- 腾讯营销洞察 (TMI) × BCG 《中国奢侈品市场洞察报告 (2024) 》 ³³
 - 新华社转载北京商报 《奢侈品旧包改造不止一眼假》 ³²
 - 得物 (得物App/应用商店说明) “先鉴别，后发货”相关描述 ³⁵

B2) What transfers from PSA to HSA, and what must change (with plans)

Transferable components mapping (PSA mechanic → HSA analog → plan → risk/constraint)

PSA mechanic	HSA analog	Implementation plan	Risk / constraint
Numeric grade as price language	Hermès-condition grade + originality grade	Publish rubric; train graders; require multi-rater agreement; build “grade notes” fields	Condition subjectivity → disputes; requires reliability audits (must be measured)
Multi-grader review	Two-person grading + QC adjudication	Mirror PSA claim: at least two graders must agree; third-party audit sample weekly	Labor cost ↑ ; must prevent “rubber stamping” (audit)
Financial guarantee + correction	HSA financial guarantee tied to declared value	Offer capped guarantee; reserve policy; explicit dispute workflow	Tail risk severe (bag values high); guarantee must be priced and limited
Tamper-evident trust artifact	“Digital slab” seal kit + cryptographic NFC	Use cryptographic NFC + tamper-evident attachment; bind to photo evidence	Tag decoupling/transplant attack; UX constraints for luxury buyers
Registry + cert verification	Public certificate verification + evidence hash	Cert lookup + NFC challenge–response verification endpoint	Privacy + data exposure tradeoffs; must avoid leaking serials useful to counterfeiters

PSA mechanic	HSA analog	Implementation plan	Risk / constraint
Population report	HSA “bag census”	Aggregate submissions by model/leather/year/region; anonymize	Data quality; brand pressure; requires careful disclosure policies

Non-transferable (must redesign) - Bags are repairable and modifiable. HSA must define *state models* (validity depends on custody and subsequent interventions). PSA can rely on slab integrity; HSA cannot.

- **Physical “slab” isn’t viable.** A bag slab is socially unacceptable and impractical; instead, the “artifact” must be a minimal, non-damaging seal + digital record.

- **Claim severity is larger.** A counterfeit graded “authentic” can be a \$10k–\$200k event; HSA must be more conservative than PSA in initial guarantee scope and must embrace “inconclusive.”

Mandatory: PSA slab equivalent for bags (“Digital Slab” options and evaluation)

Option set 1) Certificate-only (paper + QR) — *Rejected*. Certificate substitution and post-auth tampering are trivial. Certilogo explicitly warns that finding a code online or via photo “does not guarantee” authenticity; you must have the product to check it. ³⁷

2) Naïve NFC hangtag (static UID / static URL) — *Rejected*. Academic work notes NFC Forum lacks a tag authentication mechanism; in that absence, NFC tags cannot provide secure product identification, and cloning becomes central. ⁹

3) Cryptographic NFC + tamper-evident physical attachment (“Digital Slab” recommended) — *Proposed HSA baseline*. NXP describes AES-based SUN messages where each tap yields a unique message, verified by the backend host. ¹⁰

4) In-custody vault-only validity — *Complement, not replacement*. Works for marketplaces who can keep custody (akin to vault models), but HSA must also support out-of-custody verification if it wants to be a category standard (not just a warehouse standard).

NFC / tagging requirements (deep, threat-modeled)

Threat model (minimum set) - Tag cloning (copy identifier)

- UID copying / emulation
- Replay attacks (capture and replay a “valid” scan)
- Certificate substitution (swap cert for a different authentic bag)
- Tag removal/reattachment
- Adhesive replacement (fake seal)
- “Donor bag” tag transplant (move tag to counterfeit bag)
- Backend API abuse (scrape cert IDs; brute-force verification endpoints)

Why naïve NFC fails (plain English, source-backed) - NFC Forum tags can carry NDEF data but NFC Forum does not provide a native authentication mechanism; without cryptographic authentication, secure product identification is not achievable. ⁹

- Brands and authentication systems warn that codes seen online/photos are insufficient: a code can be copied, and verification must be tied to physical possession and secure checks. ³⁷

Minimum viable secure design (HSA recommended baseline) - Use **cryptographic NFC** with **challenge-response** or **tap-unique cryptographic message** generation. NXP’s NTAG 424 DNA describes AES-128 SUN

message generation per tap, with host-side verification. ¹⁰

- Prefer tags that support **seal-break / tamper state**. NXP describes “TagTamper” variants adding “seal-break detection.” ³⁸

- Use **tamper loops** (physical break detection). Avery Dennison’s “Tamper Loop” describes closed/open states and smartphone-checkable tamper status. ³⁹

- Require **server-side verification** with rate limits and abuse monitoring; never treat locally read tag data as proof. (Design inference driven by SUN host-verification model.) ⁴⁰

Key management and manufacturing assumptions (explicit) - Per-tag keys: each tag has diversified secrets; compromise of one tag does not compromise fleet (estimate; required best practice).

- **Provisioning:** keys injected at manufacturing or in HSA secure facility; keys stored in HSM; audit logs immutable (estimate).

- **Revocation/rotation:** backend maintains revocation list; tags can be marked “invalid” if suspected transplant or compromise (estimate).

- **If key leaks:** revoke affected batch; rotate backend; force re-sealing for “green” status (estimate).

UX and legitimacy constraints - Hermès buyers will reject anything that looks like a cheap hangtag. The seal must be discreet, non-damaging, and consistent with luxury aesthetics. (Unknown; must be tested with dealers/HNW buyers.)

Validity states (critical to preventing “post-auth drift”) - Certified-in-lab: authenticity + grade determined; tag bound; evidence captured.

- **Sealed-in-custody:** item held in HSA custody; strongest claim.

- **Verified-at-checkout:** real-time NFC check during transaction; confirms tag + backend status = valid.

- **Out-of-custody (time-bound):** authenticity/grade reflect inspection time; after a time window or after certain events, verification returns “valid-at-date” not “currently guaranteed.” (Design choice; must be explicit.)

- **Re-auth required** after spa/repair, hardware replacement, dyeing, or seal damage.

B3) Comparables & adjacent attempts (why hasn’t this become standardized already?)

Below are **seven** comparables; each is analyzed only on **authenticity + rating** (not marketplace GMV).

StockX (sneakers/streetwear/handbags) - Trust mechanism (promise language): “Verified by StockX” and not endorsed by brands; handbags must be brand new. ⁶

- Rating approach: largely **binary gate** (authentic + new condition), not a nuanced grade. ⁴¹

- Dispute workflow: Buyer Promise with a short claim window (10 days) and tag-intact requirement. ⁷

- Vulnerability/failure mode: legal and reputational exposure when fakes slip through; court order found liability for distributing counterfeit goods in the Nike case (37 total pairs). ⁸

- Lesson for HSA: if your claim is “verified authentic,” adversaries (and courts) will test it. HSA must design with conservative states (“inconclusive”) and tight chain-of-custody artifacts.

GOAT (sneakers) - Trust mechanism: GOAT states products are “verified by a variety of means” including digital authentication, in-hand verification, and/or ML; crucially it acknowledges some products may be verified without physical examination. ⁴²

- Rating approach: primarily authentication/condition-as-described, not a PSA-like grade (varies by product

class).

- Scalability constraint: mixing non-physical verification can reduce cost but may weaken trust for high-value categories (HSA should avoid this for Hermès). ⁴²
- Lesson: HSA should not dilute its core claim by allowing “digital-only” verification for Hermès; Hermès needs in-hand inspection + evidence capture.

eBay Authenticity Guarantee (handbags/watches/jewelry/cards) - Trust mechanism: explicitly independent from and not endorsed by brands; authenticators verify authenticity and that condition matches listing. ⁴³

- Rating approach: generally **matching-to-listing** vs publishing a universal condition grade; and eBay notes situations where authenticity cannot be verified but condition can. ³⁶
- Monetization: bundled trust layer in marketplace; backed by eBay Money Back Guarantee depending on eligibility. ⁴⁴
- Lesson: marketplace-layer authentication tends to collapse into a dispute process rather than becoming a canonical grade standard. HSA's opportunity is to be *category standard* independent of any one marketplace.

Entrupy (device + AI; luxury authentication) - Trust mechanism: collects microscopic images and compares against a database containing “millions” of records from known authentic and counterfeit items (Entrupy description). ⁴⁵

- Pricing: Hermès Premium priced at \$119 per authentication (as listed). ⁵
- Guarantee scope: one-year financial guarantee to Entrupy's direct customers; non-transferable. ⁴⁶
- Vulnerability/failure mode: certificates are not transferable; this prevents standardization as a widely traded consumer-facing grade signal unless marketplaces integrate at the business level. ⁴⁶
- Lesson: HSA should consider similar non-transferable guarantee constraints early, but must still create a transferable *verification* artifact (even if guarantee is limited to direct customers).

Chrono24 Certified (watches) - Trust mechanism: certified watches checked by a verified watchmaker; positioned as the basis of an authenticity guarantee. ⁴⁷

- Artifact: certificate of authenticity + authenticity guarantee. ⁴⁸
- Lesson: watch market shows that third-party “verified” can scale when routed through professionals (watchmakers) and anchored in documented criteria; HSA needs similarly explicit criteria and professional-grade auditability.

Certilogo (digital code verification for brands) - Trust mechanism: code-based authenticity check; but warns that finding a code online/photo does not guarantee authenticity — you must have the product. ³⁷

- Lesson: codes alone are not chain-of-custody. HSA must design against code reuse and certificate substitution.

China-specific: 得物 (Dewu/Poizon) - Trust mechanism: “鉴别服务” and “先鉴别，后发货,” multi-step checks; app-store description claims platform has authenticated “超4000万件” items (self-reported). ³⁵

- Rating approach: primarily authenticity and defect screening, not a universal grade signal transferable across the ecosystem.
- Lesson: China demonstrates massive demand for process-based trust, but also shows that platform-native authentication doesn't automatically become an independent standard. HSA can become the cross-market “PSA-like” layer for Hermès if it publishes a stable grading language and verification rail.

Mandatory: PSA vs StockX vs HSA (authenticity + rating only)

Dimension	PSA	StockX	HSA (proposed)
Object type + environment	Cards/items encapsulated; lab grading; slab is long-lived	Marketplace intermediary; focuses on “new” condition for handbags	Hermès bags; lab inspection + evidence capture + seal kit
Authentication claim	PSA guarantees grading/auth under its standards; buyback/refund if grade/auth issue later determined	“Verified by StockX” (not brand-endorsed); Buyer Promise dispute process	“Independent expert opinion” with explicit validity states; conservative “inconclusive” option
Rating/grade signal	Numeric 1–10 grading standard; qualifiers/no-grade taxonomy	Mostly binary gate: authentic + new condition	Dual signal: authenticity verdict + condition grade + originality state
Trust artifact	PSA holder (tamper-evident), cert number, public cert verification	StockX tag + platform records; claim requires tag attached	“Digital Slab”: tamper-evident loop + cryptographic NFC; public cert verification + evidence hashes
Liability/ guarantee scope	Buyback/refund difference in value under stated conditions (PSA Financial Guarantee) ¹	Buyer Promise (refund/replace if incorrect); strict claim timing/conditions ⁴⁹	Capped guarantee tied to declared value; reserve pricing; structured dispute flow (estimate)
Incentives / conflicts	Revenue from submissions; risk is grade inflation; mitigated by policy + reputation	Revenue from transaction fees; conflict: approve = higher conversion; court-tested risk (Nike case) ⁸	Must separate grading comp from transaction outcomes; ban pay-per-pass incentives (design requirement)
Scaling bottleneck	Labor + QA; backlog risk documented	Authentication labor + logistics; dispute load	Labor + training + evidence capture + secure tag ops; guarantee tail risk
Network effects source	Registry + population report + marketplace quoting in PSA grades ⁵⁰	Two-sided marketplace liquidity + pricing graph	Cross-market API rail + dealer adoption + “bag census” + standardized grade language

Key implications for HSA design

HSA must behave more like PSA than StockX: the defensible moat is the grade language + verification registry + financial accountability. But unlike PSA, HSA must explicitly model post-certification state changes

and incorporate cryptographic tag security; otherwise it will become a “certificate vendor” rather than a standard.

Part C — Proposal: what needs to be done

C1) Product + service lineup (MVP → platform; 3-5 tiers, trust-first)

Tier 1: HSA Authenticate (A) — in-hand authenticity verdict - Customer receives: digital certificate + public verification page; high-resolution evidence set (select views), no physical tag (optional add-on).

- Trust claim language: “Independent third-party opinion; not affiliated with or endorsed by Hermès.” (Benchmark: eBay explicitly says its authenticity services are not endorsed by any brands sold on eBay.) ³⁰

- Workflow: intake → standardized photo/evidence capture → primary authenticator → secondary authenticator → adjudication if disagreement → cert issued (pass / fail / inconclusive).

- Guarantee: limited guarantee only for direct submitter; caps based on declared value (design inspired by Entrupy’s direct-customer, time-bounded guarantee — but HSA should choose its own terms). ⁴⁶

- Pricing: **\$250–\$450** (estimate; depends on model/value/turnaround).

- API rail: verification endpoint returns cert validity + status (valid / revoked / expired / reauth-required).

Tier 2: HSA Grade + Report (A+G) — authenticity + condition grade + originality state - Customer receives: digital certificate + structured grading report (condition sub-scores) + public verification page.

- Trust claim: condition grade is a standardized rubric; authenticity is opinion at time of inspection; post-inspection changes can invalidate grade.

- Workflow additions: condition grading rubric + originality assessment (“Original / Repaired / Modified / Unknown”).

- Dispute process: formal appeals window; re-review by senior adjudicator; publish SLA.

- Pricing: **\$550–\$850** (estimate; see C3).

Tier 3: HSA Digital Slab Seal (A+G+S) — cryptographic NFC + tamper-evident seal kit - Customer receives: (i) physical seal kit attached to bag (non-damaging), (ii) cryptographic NFC tag bound to certificate, (iii) verification UX (“tap to verify”).

- Security design: AES-based tap-unique message tags (e.g., NTAG 424 DNA class design) with backend verification; optional TagTamper or tamper-loop hardware. ⁵¹

- Validity state model: “sealed-in-custody” vs “out-of-custody,” with reauth triggers explicitly enumerated.

Tier 4: HSA Vault-Custody Certification (A+G+S+CoC) - Customer receives: everything above + custody state (“in vault”) with stronger guarantee language because the item cannot be altered without detection.

- Analog: PSA Vault terms underscore inspection/reporting deadlines; vault models enable stronger claims because custody is controlled. ⁵²

- Monetization: storage fee + verification fees; platform partners get higher-confidence inventory.

Tier 5: HSA Re-Verification (ReAuth / Post-Spa) - Designed for events that break validity: spa/repair, seal damage, ownership transfer.

- Analog: PSA Reholder authenticates holder/card before reholding and warns that cracked holders

trigger raw examination to ensure original grade/auth still valid. 17

- HSA must treat any seal break as requiring re-verification.

C2) 12-18 month rollout plan

MVP scope (first 90 days):

Ship Tier 1 + Tier 2 with a narrow Hermès subset (e.g., Birkin/Kelly/Constance in common leathers, exclude exotic initially) and *do not ship a seal/tag* until the grading rubric stabilizes (to avoid scaling a flawed artifact). This is consistent with the “trust-first, artifact-second” principle implied by NFC security literature: without strong authentication design, tags are not secure identifiers. 53

Operational ramp (months 3-12):

- Build lab SOP: intake imaging stations; controlled lighting; measurement tools; hardware/engraving validation; materials reference library.
- Hiring: target ex-resale authenticators, luxury repair specialists, leather/hardware experts; define training curriculum and certification tests.
- QA cadence: weekly blind regrades; monthly external audit panel; publish inter-rater reliability metrics internally.

Data flywheel (months 1-18):

- Ground truth creation: acquire authentic exemplars and known counterfeit exemplars (legally) for training/reference.
- Evidence schema: require consistent photo angles + macro shots of stamps/hardware/stitching; ensure evidence is good enough to support disputes.
- Counterfeit library: versioned; tag patterns; known failure patterns.

Quarterly KPIs (trust-centric)

- Throughput/day/authenticator; turnaround time distribution; rework rate; appeal rate; dispute rate; reserve utilization; partner adoption (listings using HSA grade language); API verification calls; percent of verifications returning “valid & sealed” vs “valid-at-date.”

C3) Unit economics & capacity model (explicit assumptions; sensitivity)

Bottom-up time model (per bag; Tier 2 Grade + Report baseline)
(All **Estimated**; must be measured with time studies in pilot.)

Step	Minutes (E)	Role (E)	Notes
Intake + chain-of-custody logging	10	Ops	barcode, weigh, photo of packaging
Standardized photography (macro + full)	20	Tech	critical for disputes
Authenticity exam (primary)	35	Authenticator	stamp/hardware/material logic
Authenticity exam (secondary)	25	Authenticator	independent second look

Step	Minutes (E)	Role (E)	Notes
Condition grading + sub-score entry	25	Senior grader	rubric-based
Adjudication (only if disagreement)	0–20	Lead	assume 15% of cases
Final QC + cert issuance	10	QC	formatting + sanity check
Packaging/return	10	Ops	

Expected labor minutes: ~135 minutes per bag (E), plus adjudication tail.

Wage + overhead assumptions (illustrative) - Authenticator fully loaded: **\$40–\$70/hr** (E; geography-sensitive)

- Ops/QC fully loaded: **\$22–\$35/hr** (E)

- Facility/security/insurance allocation: **\$25–\$60/bag** (E; volume-sensitive)

- Dispute reserve: **\$10–\$40/bag** (E; must be calibrated from observed loss)

Illustrative cost per bag (Tier 2 A+G) - Labor: ~2.25 hrs blended at ~\$45/hr = **\$101** (E)

- Overhead allocation: **\$40** (E)

- Materials (no seal): **\$5–\$15** (E)

- Reserve: **\$20** (E)

Total COGS estimate: ~\$166–\$220/bag (E)

Tier 3 with “Digital Slab” seal kit (A+G+S) Add: - Secure tag + tamper loop + packaging: **\$10–\$35/bag** (E; depends on chip + assembly)

- Extra labor for attachment + verification: **10–15 min** (E)

Total COGS estimate: ~\$190–\$320/bag (E)

Pricing required for target gross margin (sensitivity) - If COGS = \$220 and target GM = 60%, price ≈ **\$550**.

- If COGS = \$320 and target GM = 60%, price ≈ **\$800**.

(These are mechanistically consistent with PSA's demonstrated ~57% FY2020 gross margin ceiling as an analog, but HSA has higher liability severity; therefore HSA should target **higher dollar contribution per unit** than PSA. ²³)

Break-even volume (single hub; estimate) Assume fixed costs (rent, security, management, tooling, software, compliance) **\$150k–\$350k/month** (E).

If contribution margin per bag is **\$350** (price \$650 – COGS \$300), break-even is ~**430–1,000 bags/month** (E).

This is directionally consistent with draft claims of ~800/month, but HSA must validate with actual time studies and wage geography. (Drafts cited in appendix.) filecite turn0file1 filecite turn0file0

Fraud/chargeback/dispute reserve logic (how to compute, not guess) Reserve per bag should be computed as:

Reserve = P(false negative) × E(loss severity | false negative) + P(dispute) × E(cost per dispute)

Where loss severity is bounded by declared value caps and policy language (see legal posture).

C4) Growth trajectory + exit scenarios (trust-standard adoption, not GMV)

3–5 year trajectory (trust milestones) - Year 1: establish rubric + reliability; 2–3 dealer partners; ship Tier 1–2; demonstrate low dispute/loss ratio.

- Year 2: ship Tier 3 (secure seal kit) once attack-tested; integrate verification API with 2–3 resale platforms; publish “bag census” aggregates.

- Year 3: expand to broader luxury categories (selectively) or deepen Hermès moat via richer authenticity corpus; begin vault-custody tier at scale.

- Years 4–5: become de facto standard in Hermès resale listings for top dealers; attach rate to verification API increases; insurance partnerships feasible if loss ratio stable.

Exit pathways - Strategic: large resale platforms (seeking independent standard), luxury-adjacent trust infra, insurers/underwriters interested in authenticated inventories, or authentication incumbents (device + data consolidation, e.g., Entrupy-like players).

- Roll-up logic: combine with repair/spa networks only if chain-of-custody and independence can be preserved (avoid conflicts).

- IPO plausibility: only if HSA becomes multi-category standard with stable loss ratio and high-margin recurring rails (verification API + custody services).

Exit-ready milestones - Documented inter-rater reliability; controlled loss ratio; audited tag security; platform integrations where HSA verification becomes default checkout behavior (analog: PSA + eBay integration). ⁵⁴

Adversarial threat model, risk register, falsifiable hypotheses

Adversarial threat model (bags; minimum 12 vectors, with controls)

This table is intentionally *operator-centric* (what we do, how it can fail).

Fraud vector	Control(s)	Detection method	Residual risk
Component swapping (hardware/lock/feet)	Originality grading + part-level inspection; photo corpus	Macro + weight + hardware markings	Medium (high-quality swaps)
Hardware replacement (after auth)	Seal kit on attachment points; reauth trigger after seal break	NFC status + physical loop/tamper evidence	Medium
Stamp manipulation	Multi-angle macro imaging; stamp logic rules	Expert review + anomaly flags	Medium-high
Receipt/box/dustbag laundering	Do not treat accessories as authenticity proof; optional accessory grading	Accessory authenticity separate from bag	Medium

Fraud vector	Control(s)	Detection method	Residual risk
Certificate substitution	Cryptographically bind cert ↔ tag ↔ evidence set; public verification	Verification endpoint mismatch blocks	Low if crypto + backend correct
Tag cloning	Cryptographic NFC (tap-unique message)	Backend challenge verification	Low if keys protected 40
Tag transplant ("donor bag")	Tamper-evident loop + destructive attachment; "seal broken" state	Tamper loop open state + invalidation	Medium (must be well-designed) 55
Adhesive replacement	Avoid adhesive-only as primary security	Seal design is mechanical + tamper loop	Low-medium
Repair/spa obfuscation	Originality state taxonomy; mandatory disclosure; reauth required	Evidence comparison at reauth	Medium
Odor concealment (smoke/mold)	Standard sensory check + UV + documentation	Condition sub-score + notes	Medium
Leather re-dye / recolor	UV + microscopy where needed; rubric flags	Condition/originality classification	Medium-high
"Return fraud" (swap after authenticity)	Seal required for refunds; custody logs	Tag status + evidence at intake	Medium
Shipping intercept / swap in transit	Tamper-evident inbound packaging; chain-of-custody logging	Package photo + weight variance	Medium
Counterfeit designed to pass HSA rubric	Red-team updates; frequent reference library updates	Drift monitoring; known-fake regression tests	Always non-zero

Risk register (minimum 10 obstacles; severity, mitigation, early warning, kill criteria)

Obstacle	Severity	Mitigation	Early warning	Kill criteria
Brand/IP litigation (injunction risk)	High	Conservative claims; nominative fair use hygiene; avoid implying affiliation	C&D letters; platform partner anxiety	Counsel rates existential risk "high"
Grade inconsistency (no standard forms)	High	Training + multi-rater + audits; publish rubric	Appeals spike; partner complaints	<75% within ±1 notch at 60 days

Obstacle	Severity	Mitigation	Early warning	Kill criteria
Guarantee tail loss blows up	High	Cap by declared value; reserve; conservative “inconclusive”	Reserve burn >1% declared value	Loss ratio >1% declared value in pilot
Tag decoupling discovered (public exploit)	High	Cryptographic NFC + tamper loop; bounty program	Exploit reports; mismatch incidents	Any confirmed exploit yields “valid”
Talent scarcity (true experts)	Medium	Apprentice pipeline; compensation; SOP encoding	Throughput stalls; quality drift	Cannot staff to run-rate by month 4
Counterfeit adaptation	Medium	Red team; counterfeit library; continuous updates	Rising inconclusive/false negatives	Seeded test fail (≥ 1 miss)
Regulatory consumer-protection action	Medium	Clear disclaimers; fair dispute handling; documented processes	Complaint volume; regulator inquiry	Mandatory changes destroy economics
Partner adoption fails (no distribution)	High	Dealer program; integration; ROI tests	<3 serious LOIs in 60 days	No measurable premium in A/B test
Data leak (cert scraping)	Medium	Rate limits; tokenized verification; abuse monitoring	Suspicious API traffic	Persistent abuse unmitigated
Reputational scandal (one big miss)	High	Conservative cert states; incident response	Viral counterfeit incident	Standard collapses (partners abandon)

Falsifiable hypotheses (minimum 8; each with sample size, metric, threshold, time window)

1) H1: Hermès dealers will pay for a *condition grade*, not just authenticity.

Sample: n=20 dealers; Test: pricing survey + signed LOIs; Pass: ≥ 8 commit to list HSA grade in listings within 60 days; Fail: <4.

2) H2: HSA grade increases liquidity/price.

Sample: n=200 listings A/B; Metric: conversion rate or net realized price; Pass: $\geq 10\%$ higher conversion or $\geq 5\%$ higher price; Window: 60 days.

3) H3: Inter-rater reliability is achievable at PSA-like levels.

Sample: n=120 graded bags; Metric: agreement rates; Pass: $\geq 85\%$ within ± 1 notch; Window: 45 days.

4) H4: HSA can keep false negatives below existential thresholds.

Sample: n=40 seeded counterfeits; Metric: counterfeit passes as authentic; Pass: 0; Window: 30 days.

5) **H5: Secure tag system resists practical attacks.**

Sample: n=30 tags; Attack: clone/replay/transplant; Pass: 0 successful validations; Window: 30 days.

6) **H6: Dispute process can be fast enough to preserve trust.**

Sample: ≥ 30 real disputes or simulated; Metric: median resolution time; Pass: ≤ 10 business days; Window: 60 days.

7) **H7: “Inconclusive” state reduces loss ratio without killing adoption.**

Sample: n=300 submissions; Metric: loss ratio + customer satisfaction; Pass: loss ratio $\leq 0.5\%$ declared value and NPS ≥ 30 ; Window: 90 days.

8) **H8: API verification rail will be used by platforms if SLA is reliable.**

Sample: n=3 platform partners; Metric: verification calls/day + uptime; Pass: $\geq 10k$ calls/month and $\geq 99.9\%$ uptime for 30 days; Window: 60 days.

Minimum legal/claims posture

Not legal advice. Operating posture + benchmark language (official sources).

Core posture (“independent opinion,” not brand-endorsed) - Benchmark: eBay states Authenticity Guarantee “is not affiliated with or endorsed by any of the brands sold on eBay.” ³⁰

- Benchmark: StockX emphasizes “Verified by StockX” and “not endorsed by any brands.” ⁵⁶

HSA should adopt similar language prominently and consistently, including in the verification UI.

Define exactly what is being promised HSA must separate: 1) **Authentication opinion** at time of inspection

2) **Condition/originality grade** at time of inspection

3) **Seal validity state** (in-custody vs out-of-custody)

4) **Guarantee scope** (who is covered, for how long, and what events void it)

This is not theory: StockX’s Buyer Promise ties claims to time limits and the presence of an attached tag, effectively conditioning the promise on chain-of-custody artifacts. ⁷

Entrupy similarly limits its guarantee: valid to direct customers for one year and not transferable. ⁴⁶

Why brand/IP risk is real in resale Litigation in resale often contests marketing language and implied affiliation; courts scrutinize representations. The Chanel v. WGACA litigation (publicly reported) shows brands pursue claims including false advertising and trademark issues; such cases raise the bar for how resellers/authenticators talk about brands. ⁵⁷

HSA must assume Hermès could test boundaries similarly.

Practical HSA claims guidelines - Avoid “Hermès-certified” or any suggestion of partnership.

- Avoid unqualified “100% authentic” claims. Courts and plaintiffs will frame such language as a guarantee that can be falsified (as discussed in Nike v StockX context, where “100% Verified Authentic” language was at issue). ⁵⁸

- Use structured, testable language: “HSA Certified Authentic (in-hand, multi-review, evidence archived).”

- Provide clear consumer remedy + SLA; align with consumer protection expectations (refund/repurchase mechanics, clear timelines).

Appendix — Phase 0 critique of the three drafts, plus “IC message” and final falsification checklist

Phase 0 — Read and critique the 3 drafts (mandatory)

Draft references (attachments) - Draft A: “HSA Investment Memo_ Building the PSA of Hermès Bags”
filecite turn0file2

- Draft B: “Gemini 1” filecite turn0file1

- Draft C: “gemini 2” filecite turn0file0

Draft-to-Requirement Coverage Matrix

Requirement section	Draft A coverage / quality	Draft B coverage / quality	Draft C coverage / quality	What is missing or wrong (summary)
PSA case study (history, mechanisms)	Med / Med	Med / Low	Med / Low	Lacked SEC-grounded mechanism detail; did not extract cost drivers correctly; insufficient linkage between backlog/tiers and economics (fixed vs variable).
PSA unit economics from SEC	Med / Low	Low / Low	Low / Low	Used headline margins without fully grounding cost components; per-item economics often asserted without SEC support. Corrected in report via SEC 10-K/10-Q. ²³
StockX promise language + rating approach	Med / Med	Low / Low	Low / Low	Drafts did not quote exact constraints (10-day window; tag must remain attached; “new only” for handbags). Corrected with StockX help pages and Buyer Promise. ⁵⁹
NFC/secure tagging threat model	Low / Low	Low / Low	Med / Low	Drafts mentioned NFC but did not threat-model cloning/replay/transplant or key management. Corrected with NFC security literature + NXP SUN architecture + tamper loops. ⁶⁰

Requirement section	Draft A coverage / quality	Draft B coverage / quality	Draft C coverage / quality	What is missing or wrong (summary)
China: credible Chinese-language evidence	Low / Low	Low / Low	Low / Low	Lacked credible Chinese-language sources and regulatory context; added Tencent×BCG report, Xinhua reporting, and Dewu platform descriptions + Reuters context. ⁶¹
Rating rubric design + auditability	Med / Low	Med / Low	Med / Low	Drafts proposed grades but did not design inter-rater reliability systems, audit loops, or spa/repair taxonomy. Rebuilt in Part B/C + hypotheses.
Legal posture	Med / Low	Low / Low	Low / Low	Drafts referenced IP risk but did not benchmark exact disclaimer language. Corrected with eBay/StockX official disclaimers and court-tested context. ⁶²
Falsification tests	Med / Med	Med / Low	Med / Low	Drafts had checklists but not always with sample sizes and pass/fail. Rebuilt with explicit thresholds (Exec + Threat/Hypotheses).

Error Log and “Questionable Claims” Register

Below are the most material questionable claims from the drafts, with disposition. (Not exhaustive of every minor number; exhaustive focus is on *decision-grade* claims.)

Draft claim (paraphrased)	Why questionable	Fix approach
“PSA 65M+ items certified”	PSA’s current homepage markets “Over 80 Million Collectibles Authenticated & Graded,” suggesting draft number is stale/outdated. ⁶³	Fix by primary source: use PSA homepage as current claim; label SEC-era counts as historical. ⁶⁴
“PSA graded ~19.26M items in 2025; ~72% share” (via GemRate/CLLCT)	Not primary; relies on secondary tracking and could be biased/partial	Reframe as estimate or remove from core IC thesis unless validated with primary datasets.

Draft claim (paraphrased)	Why questionable	Fix approach
"FY2020 trading card grading fees averaged \$10.27/card; ranged \$1–\$5,000"	Not supported by SEC filings; PSA current tier prices are materially higher than \$10, and historical averages aren't disclosed in SEC filings	Remove as "verified"; treat pricing via PSA published service tiers as primary. ⁶⁵
"Expected guarantee loss ~0.18% of declared value"	An assumption, not fact; depends on counterfeit base rate and loss severity	Reframe as estimate with sensitivity; require pilot loss-ratio test (see falsification).
"Breakeven ~800 bags/month"	Unsupported without transparent labor minutes + wage + fixed cost model	Fix by explicit capacity model (C3) and treat as estimate with sensitivity.
Generic "NFC tag" recommendations	Fails to explain why static identifiers can be cloned/replayed; no key management; no state model	Fix by primary security architecture: cryptographic NFC SUN + tamper loop + backend verification. ⁶⁰
StockX "trust" described without exact promise constraints	Promise is conditional (10-day claim window; tag attached); also court-tested fakes issue is central	Fix by official language + court order. ⁶⁶
China market points without reputable Chinese-language sourcing	Violates source hygiene; requires Chinese-language evidence	Fix by Chinese-language reports + official platform language (see Part B). ⁶⁷

Missing-Insight Map (what the drafts failed to do)

- **PSA economics (SEC-level causal drivers):** drafts didn't clearly tie margins to cost categories (labor, occupancy/security/insurance, warranty) and demand throttles (turnaround pricing). Corrected using Collectors Universe SEC disclosures. ²⁶
- **StockX exact claim language + controversy impact:** drafts underweighted how the *court record* (Nike v StockX) functions as an external falsification of "verification" narratives. Corrected with the Mar 4, 2025 order and StockX's own statement. ⁶⁸
- **NFC/tamper-evident threat model:** drafts treated NFC as a feature, not an adversarial system. Corrected using NFC authentication literature + NXP SUN architecture + tamper loops. ⁶⁰
- **China-specific ecosystem and consumer protection context:** drafts lacked reputable Chinese-language sources; corrected with Tencent×BCG report, Xinhua reporting on bag modification markets, and Dewu's "authenticate-first" process descriptions (self-reported). ⁶⁷
- **Rating rubric auditability + spa/repair handling:** drafts proposed "grades" without building reliability measurement and re-auth triggers. Corrected with explicit reliability tests, state models, and product tiers.

Primary-source link pack (direct links; accessed 2026-01-27 Asia/Taipei)

(Per instruction: URLs are provided in code format. Access date for all: 2026-01-27.)

SEC (Collectors Universe) FY2020 10-K:

<https://www.sec.gov/Archives/edgar/data/1089143/000149315220016754/form10-k.htm>

SEC (Collectors Universe) FY2021 Q1 10-Q (revenue recognition, cost drivers, backlog):

<https://www.sec.gov/Archives/edgar/data/1089143/000149315220020321/form10-q.htm>

PSA Pricing (grading service levels):

<https://www.psacard.com/services/tradingcardgrading>

PSA FAQ (financial guarantee summary; multi-grader agreement; regrades; no-grade taxonomy):

<https://www.psacard.com/support/faq>

PSA Security buyer guide (tamper evidence / sonic weld):

<https://www.psacard.com/services/psasecurityabuyersguide>

PSA Population Report:

<https://www.psacard.com/pop>

PSA Set Registry history:

<https://www.psacard.com/articles/articleview/9374/the-history-of-the-psa-set-registry>

StockX Our Process (not brand-endorsed; handbags new-only language):

<https://stockx.com/about/our-process>

StockX Buyer Promise:

<https://stockx.com/help/articles/StockX-Buyer-Promise>

Court order (Nike v StockX; Mar 4 2025 order text republished):

<https://www.casemine.com/judgement/us/67c924d52016e547bca7c749>

Entrupy Hermes pricing:

<https://www.entrupy.com/hermes-pricing/>

Entrupy Luxury Financial Guarantee (non-transferable; 1-year):

<https://www.entrupy.com/guarantee/luxury/>

Entrupy How-to-Use (microscopy + ML; millions of records):

<https://www.entrupy.com/how-to-use/>

NXP NTAG 424 DNA datasheet (SUN; AES-128; host verification):

<https://www.nxp.com/docs/en/data-sheet/NT4H2421Gx.pdf>

NXP Application Note AN12196 (Secure Dynamic Messaging):

<https://www.nxp.com/docs/en/application-note/AN12196.pdf>

Avery Dennison Tamper Loop (digital seal):

<https://rfid.averydennison.com/en/home/product-finder/circus-tamper-loop.html>

Academic paper (NFC tag authentication limitations; cloning concerns):

[https://pure.royalholloway.ac.uk/ws/files/23032289/](https://pure.royalholloway.ac.uk/ws/files/23032289/Off_Line_NFC_Tag_Authentication.pdf)

[Off_Line_NFC_Tag_Authentication.pdf](https://pure.royalholloway.ac.uk/ws/files/23032289/Off_Line_NFC_Tag_Authentication.pdf)

Dewu (得物) official site and Apple App Store description (authenticate-first workflow; self-reported volume):

<https://www.dewu.com/>

[https://apps.apple.com/tw/app/%E5%BE%97%E7%89%A9-](https://apps.apple.com/tw/app/%E5%BE%97%E7%89%A9-%E5%BE%97%E5%88%B0%E7%BE%8E%E5%A5%BD%E4%BA%8B%E7%89%A9/id1012871328?l=en)

[%E5%BE%97%E5%88%B0%E7%BE%8E%E5%A5%BD%E4%BA%8B%E7%89%A9/id1012871328?l=en](https://apps.apple.com/tw/app/%E5%BE%97%E5%88%B0%E7%BE%8E%E5%A5%BD%E4%BA%8B%E7%89%A9/id1012871328?l=en)

Tencent Marketing Insight × BCG luxury report (Chinese PDF):

https://bbx-pic.gting.com/bbx/pictures/2024/48_20240906150809_485299.pdf

Xinhua / Beijing Business Daily (bag modification market; Chinese):

<https://www.news.cn/fashion/20230519/9fe7e42e3e084182a414c13f907ecb8e/c.html>

Collectors Universe take-private press release (GlobeNewswire):

<https://www.globenewswire.com/news-release/2020/11/30/2136654/0/en/Collectors-Universe-to-be-Acquired-by-Investor-Group-Led-by-Entrepreneur-and-Collector-Nat-Turner-for-Approximately-700-Million.html>

Acquisition completion (Business Wire):

<https://www.businesswire.com/news/home/20210222005284/en/Investor-Group-Led-by-Nat-Turner-Completes-Acquisition-of-Collectors-Universe>

What we would tell the IC (tight)

HSA is viable **only** if we treat it as a *trust infrastructure company* with a cryptographically bound “digital slab” and a defensible grading standard—not as a marketplace feature. PSA’s success is not magic; it’s the repeated execution of multi-rater grading, a tamper-evident artifact, public verification rails, and financial accountability that the market can price into transactions. ⁶⁹

Hermès is harder than cards because bags change state (spa/repair/parts). That means HSA must introduce **validity states** and re-auth triggers, and its seal system must resist cloning/replay/transplant. NFC done naïvely fails; cryptographic NFC + tamper loops + backend verification is table stakes. ⁶⁰

We proceed because (a) unit economics for trust services can be excellent (PSA scale shows this), ² and (b) China and global resale behavior shows escalating demand for “authenticate-first” workflows, but no

cross-market Hermès grading standard exists yet. 70

We kill the project fast if reliability, tag security, or loss ratio fails the falsification checklist.

Falsification checklist (repeat)

1. Inter-rater reliability: $n=120$; $\geq 85\%$ within ± 1 notch; fail $< 75\%$.
 2. Seeded counterfeit test: $n=40$; 0 pass as authentic; fail ≥ 1 .
 3. Tag attack test: $n=30$; 0 successful clone/replay/transplant validations; fail ≥ 1 .
 4. Cert $\leftrightarrow$ bag binding audit: $n=200$; $\geq 99.5\%$ correct identity/state; fail systemic mismatch.
 5. Dispute SLA: $\geq 90\%$ resolved ≤ 10 business days; fail median > 15 .
 6. Dealer adoption: ≥ 6 LOIs to list HSA grades; fail < 3 .
 7. Price premium/liquidity: $n=200$ A/B; $\geq 10\%$ conversion or $\geq 5\%$ price; fail no effect.
 8. Loss ratio ceiling: $\leq 0.5\%$ of declared value in pilot; fail $> 1.0\%$.
 9. Throughput: ≥ 6 graded bags/authenticator/day by week 8; fail < 4 .
 10. Brand/IP posture: no existential legal threat in first 60 days; fail if injunction risk becomes likely.
-

1 69 <https://www.psacard.com/support/faq>

<https://www.psacard.com/support/faq>

2 23 [sec.gov](https://www.sec.gov)

<https://www.sec.gov/Archives/edgar/data/1089143/000149315220016754/form10-k.htm>

3 18 26 <https://www.sec.gov/Archives/edgar/data/1089143/000149315220020321/form10-q.htm>

<https://www.sec.gov/Archives/edgar/data/1089143/000149315220020321/form10-q.htm>

4 28 65 [PSA Estimated Turnaround Time Updates](https://www.psacard.com/info/submission-updates?utm_source=chatgpt.com)

https://www.psacard.com/info/submission-updates?utm_source=chatgpt.com

5 <https://www.entrupy.com/hermes-pricing/>

<https://www.entrupy.com/hermes-pricing/>

6 56 <https://stockx.com/about/our-process/>

<https://stockx.com/about/our-process/>

7 49 59 66 <https://stockx.com/help/articles/what-is-the-stockx-buyer-promise/>

<https://stockx.com/help/articles/what-is-the-stockx-buyer-promise/>

8 68 <https://www.casemine.com/judgement/us/67c924d52016e547bca7c749>

<https://www.casemine.com/judgement/us/67c924d52016e547bca7c749>

9 53 60 https://pure.royalholloway.ac.uk/ws/files/23032289/Off_Line_NFC_Tag_Authentication.pdf

https://pure.royalholloway.ac.uk/ws/files/23032289/Off_Line_NFC_Tag_Authentication.pdf

10 40 51 <https://www.nxp.com/docs/en/data-sheet/NT4H2421Gx.pdf>

<https://www.nxp.com/docs/en/data-sheet/NT4H2421Gx.pdf>

11 <https://pages.ebay.co.uk/authenticity-guarantee-watches-seller/terms/>

<https://pages.ebay.co.uk/authenticity-guarantee-watches-seller/terms/>

12 27 54 <https://www.psacard.com/articles/articleview/15161/psa-grading-now-built-into-ebay-checkout>

<https://www.psacard.com/articles/articleview/15161/psa-grading-now-built-into-ebay-checkout>

13 <https://www.entrupy.com/search-certificate/>

<https://www.entrupy.com/search-certificate/>

14 25 <https://www.psacard.com/services/psasecurityabuyersguide>

<https://www.psacard.com/services/psasecurityabuyersguide>

15 <https://www.psacard.com/cert/103527207/psa>

<https://www.psacard.com/cert/103527207/psa>

16 50 <https://www.psacard.com/info/psa-verified>

<https://www.psacard.com/info/psa-verified>

17 31 <https://www.psacard.com/services/tradingcardgrading/reholder>

<https://www.psacard.com/services/tradingcardgrading/reholder>

19 <https://www.globenewswire.com/news-release/2020/11/30/2136654/0/en/Collectors-Universe-to-be-Acquired-by-Investor-Group-Led-by-Entrepreneur-and-Collector-Nat-Turner-for-Approximately-700-Million.html>

<https://www.globenewswire.com/news-release/2020/11/30/2136654/0/en/Collectors-Universe-to-be-Acquired-by-Investor-Group-Led-by-Entrepreneur-and-Collector-Nat-Turner-for-Approximately-700-Million.html>

- 20 <https://www.businesswire.com/news/home/20210222005284/en/Investor-Group-Led-by-Nat-Turner-Completes-Acquisition-of-Collectors-Universe>
<https://www.businesswire.com/news/home/20210222005284/en/Investor-Group-Led-by-Nat-Turner-Completes-Acquisition-of-Collectors-Universe>
- 21 <https://www.psacard.com/articles/articleview/10556/collectors-universe-acquires-trading-card-valuation-platform-ladder>
<https://www.psacard.com/articles/articleview/10556/collectors-universe-acquires-trading-card-valuation-platform-ladder>
- 22 <https://www.prnewswire.com/news-releases/collectors-and-ebay-transactions-close-summer-launch-expected-for-integrated-end-to-end-hobby-experience-302148382.html>
<https://www.prnewswire.com/news-releases/collectors-and-ebay-transactions-close-summer-launch-expected-for-integrated-end-to-end-hobby-experience-302148382.html>
- 24 <https://www.collectors.com/leadership>
<https://www.collectors.com/leadership>
- 29 <https://support.psacard.com/s/article/verify-cert>
<https://support.psacard.com/s/article/verify-cert>
- 30 43 62 <https://ebay.com/authenticity-guarantee/handbags>
<https://ebay.com/authenticity-guarantee/handbags>
- 32 <https://www.news.cn/fashion/20230519/9fe7e42e3e084182a414c13f907ecb8e/c.html>
<https://www.news.cn/fashion/20230519/9fe7e42e3e084182a414c13f907ecb8e/c.html>
- 33 61 67 https://bbx-pic.gting.com/bbx/pictures/2024/48_20240906150809_485299.pdf
https://bbx-pic.gting.com/bbx/pictures/2024/48_20240906150809_485299.pdf
- 34 70 <https://www.reuters.com/business/retail-consumer/chinas-booming-grey-markets-add-woes-luxury-brands-2024-10-21/>
<https://www.reuters.com/business/retail-consumer/chinas-booming-grey-markets-add-woes-luxury-brands-2024-10-21/>
- 35 <https://apps.apple.com/tw/app/%E5%BE%97%E7%89%A9-%E5%BE%97%E5%88%B0%E7%BE%8E%E5%A5%BD%E4%BA%8B%E7%89%A9/id1012871328>
<https://apps.apple.com/tw/app/%E5%BE%97%E7%89%A9-%E5%BE%97%E5%88%B0%E7%BE%8E%E5%A5%BD%E4%BA%8B%E7%89%A9/id1012871328>
- 36 <https://www.ebay.com/help/buying/buying-authenticity-guarantee/buying-authenticity-guarantee?id=5470>
<https://www.ebay.com/help/buying/buying-authenticity-guarantee/buying-authenticity-guarantee?id=5470>
- 37 <https://discover.certilogo.com/pages/faq-consumer>
<https://discover.certilogo.com/pages/faq-consumer>
- 38 <https://www.nxp.com/products/rfid-nfc/nfc-hf/ntag-for-tags-and-labels/ntag-424-dna-424-dna-tagtamper-advanced-security-and-privacy-for-trusted-iot-applications%3ANTAG424DNA>
<https://www.nxp.com/products/rfid-nfc/nfc-hf/ntag-for-tags-and-labels/ntag-424-dna-424-dna-tagtamper-advanced-security-and-privacy-for-trusted-iot-applications%3ANTAG424DNA>
- 39 55 <https://rfid.averydennison.com/en/home/news-insights/insights/healthcare-circus-tamper-loop-a-new-digital-seal-for-your-products.html>
<https://rfid.averydennison.com/en/home/news-insights/insights/healthcare-circus-tamper-loop-a-new-digital-seal-for-your-products.html>

- 41 <https://stockx.com/help/articles/what-is-the-condition-of-items-sold-on-stockx>
<https://stockx.com/help/articles/what-is-the-condition-of-items-sold-on-stockx>
- 42 <https://www.goat.com/verification>
<https://www.goat.com/verification>
- 44 <https://www.ebay.com/help/selling/selling-tools/ebay-authenticity-guarantee?id=4644>
<https://www.ebay.com/help/selling/selling-tools/ebay-authenticity-guarantee?id=4644>
- 45 <https://www.entrupy.com/how-to-use/>
<https://www.entrupy.com/how-to-use/>
- 46 <https://www.entrupy.com/guarantee/luxury/>
<https://www.entrupy.com/guarantee/luxury/>
- 47 <https://www.chrono24.com/certified.htm>
<https://www.chrono24.com/certified.htm>
- 48 <https://about.chrono24.com/en/press/chrono24-debuts-new-certified-program-providing-transparency-and-an-authenticity-guarantee-to-the-worlds-largest-selection-of-pre-owned-watches>
<https://about.chrono24.com/en/press/chrono24-debuts-new-certified-program-providing-transparency-and-an-authenticity-guarantee-to-the-worlds-largest-selection-of-pre-owned-watches>
- 52 <https://www.psacard.com/vaulttermsandconditions>
<https://www.psacard.com/vaulttermsandconditions>
- 57 <https://www.vogue.com/article/chanel-wins-case-against-what-goes-around-comes-around>
<https://www.vogue.com/article/chanel-wins-case-against-what-goes-around-comes-around>
- 58 <https://law.justia.com/cases/federal/district-courts/new-york/nysdce/1%3A2022cv00983/574411/250/>
<https://law.justia.com/cases/federal/district-courts/new-york/nysdce/1%3A2022cv00983/574411/250/>
- 63 64 <https://www.psacard.com/>
<https://www.psacard.com/>